IBM HR Analytics SQL Project

Business Case Study - Employee Attrition Analysis Using SQL

Dataset

IBM HR Analytics Employee Attrition Dataset (Kaggle)

Skills Used:

- SQL
- CASE WHEN
- GROUP BY
- Aggregate Functions
- Subqueries
- Window Functions (RANK)
- Business Analysis

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1. Department-wise Employee Attrition Analysis

Business Question:
Which department has the highest employee attrition percentage?

SQL Query:

```sql
-- The HR Head wants to know which department has the highest employee attrition.
SELECT Department,
       COUNT(EmployeeNumber) AS TotalEmployees,
       SUM(
       CASE
       WHEN Attrition="Yes" THEN 1
       ELSE 0
       END) AS Employees_left,
    ROUND(
        SUM(
        CASE
        WHEN Attrition ="Yes" THEN 1
        ELSE 0
        END)*100.0
        /COUNT(EmployeeNumber),
        2
        ) AS Attrition_Percentage
FROM employee_attrition
GROUP BY department
ORDER BY Attrition_Percentage DESC;
```

Query Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	TotalEmployees	Employees_left	Attrition_Percentage
Sales	446	92	20.63
Human Resources	63	12	19.05
Research & Development	961	133	13.84

Business Insight:

Sales has the highest employee attrition rate (**20.63%**).
Human Resources follows with **19.05%.**
Research & Development has the lowest attrition rate (**13.84%**).

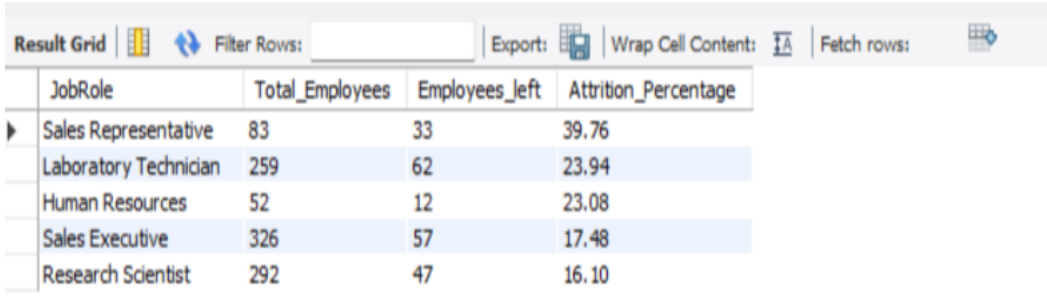
2. Job Role-wise Employee Attrition Analysis

Top 5 job roles with highest employee attrition percentage?

SQL QUERY:

```sql
-- HR wants to identify the Top 5 Job Roles with the highest attrition percentage.
SELECT JobRole,
       COUNT(employeenumber) AS Total_Employees,
       SUM(
       CASE
       WHEN Attrition="yes" THEN 1
       ELSE 0
       END) AS Employees_left,
    ROUND(
          SUM(
              CASE
              WHEN Attrition="Yes" THEN 1
              ELSE 0
              END) * 100.0
              / COUNT(employeenumber),
              2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY JobRole
ORDER BY Attrition_Percentage DESC
LIMIT 5;
```

QUERY OUTPUT:



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

JobRole	Total_Employees	Employees_left	Attrition_Percentage
Sales Representative	83	33	39.76
Laboratory Technician	259	62	23.94
Human Resources	52	12	23.08
Sales Executive	326	57	17.48
Research Scientist	292	47	16.10

Business Insight:

- **Sales Representative** has the highest attrition rate (**39.76%**).
- **Laboratory Technician** follows with **23.94%**.
- **Human Resources** records **23.08%** attrition.
- These roles may require better retention strategies and employee engagement initiatives.

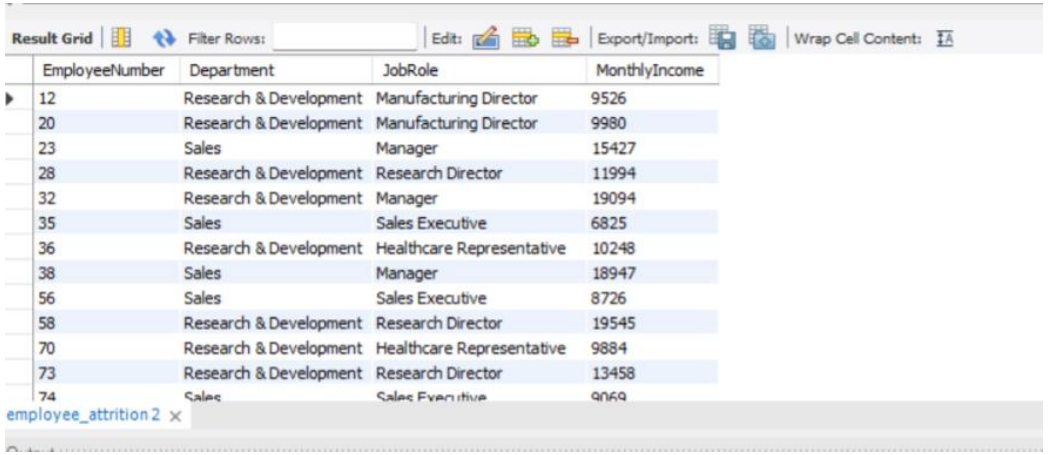
Employees Above Average Salary

Which employees earn above the company's average salary?

SQL QUERY:

```sql
-- HR wants to identify employees earning more than the company-wide average monthly income.
SELECT EmployeeNumber,
       Department,
       JobRole,
       MonthlyIncome
FROM employee_attrition
WHERE MonthlyIncome>(
SELECT AVG(MonthlyIncome)
FROM employee_attrition
);
```

QUERY OUTPUT:



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

EmployeeNumber	Department	JobRole	MonthlyIncome
12	Research & Development	Manufacturing Director	9526
20	Research & Development	Manufacturing Director	9980
23	Sales	Manager	15427
28	Research & Development	Research Director	11994
32	Research & Development	Manager	19094
35	Sales	Sales Executive	6825
36	Research & Development	Healthcare Representative	10248
38	Sales	Manager	18947
56	Sales	Sales Executive	8726
58	Research & Development	Research Director	19545
70	Research & Development	Healthcare Representative	9884
73	Research & Development	Research Director	13458
74	Sales	Sales Executive	9069

employee_attrition 2

Business Insight:

Above-average salaries are concentrated in senior and specialized roles, supporting HR in compensation benchmarking and workforce planning.

Education Field-wise Employee Attrition Analysis

Which education fields have the highest employee attrition percentage?

SQL QUERY:

```sql
-- Which education fields have the highest employee attrition percentage?
SELECT EducationField,
       COUNT(EmployeeNumber) as Total_employees,
       SUM(
       CASE
       WHEN Attrition="Yes" THEN 1
       ELSE 0
       END) AS Employees_left,
    ROUND(
          SUM(
          CASE
          WHEN Attrition = "Yes" THEN 1
          ELSE 0
          END)*100.0
          /COUNT(EmployeeNumber),
          2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY EducationField
ORDER BY Attrition_Percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EducationField	Total_employees	Employees_left	Attrition_Percentage
Human Resources	27	7	25.93
Technical Degree	132	32	24.24
Marketing	159	35	22.01
Life Sciences	606	89	14.69
Medical	464	63	13.58
Other	82	11	13.41

Business Insight: Employees from **Human Resources (25.93%)** and **Technical Degree (24.24%)** education fields have the highest attrition rates, highlighting the need for **targeted retention strategies**..

Age Group-wise Employee Attrition Analysis

How does employee attrition vary across different age groups?

SQL QUERY:

```sql
-- Which age groups have the highest employee attrition percentage?
SELECT
    CASE
      WHEN Age BETWEEN 18 AND 25 THEN "18-25"
      WHEN Age BETWEEN 26 AND 35 THEN "26-35"
      WHEN Age BETWEEN 36 AND 45 THEN "36-45"
      ELSE "56+"
    END AS Age_Group,
    COUNT(EmployeeNumber) AS Total_Employees,
    SUM(
        CASE
        WHEN Attrition = "Yes" THEN 1
        ELSE 0
        END) AS Employees_left,
    ROUND(
          SUM(
          CASE
          WHEN Attrition = "Yes" THEN 1
          ELSE 0
          END)*100.0
          /COUNT(EmployeeNumber),
          2) AS Attrition_percentage
FROM Employee_Attrition
GROUP BY Age_Group
ORDER BY Attrition_percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Age_Group	Total_Employees	Employees_left	Attrition_percentage
18-25	123	44	35.77
26-35	606	116	19.14
56+	273	34	12.45
36-45	468	43	9.19

Business Insight:

Employees aged **18–25 (35.77%)** have the highest attrition rate, indicating that younger employees are more likely to leave the organization. This highlights the need for targeted retention Strategies for early career employees.

Gender-wise Employee Attrition Analysis

How does employee attrition differ between male and female employees?

SQL QUERY :

```sql
-- How does employee attrition differ between male and female employees?
SELECT Gender,
       COUNT(EmployeeNumber) AS Total_Employees,
       SUM(
           CASE
           WHEN Attrition="Yes" THEN 1
           ELSE 0
           END) AS Employees_left,
        ROUND(
            SUM(
                CASE
                WHEN Attrition="Yes" THEN 1
                ELSE 0
                END)*100.0
                /COUNT(EmployeeNumber),
                2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY Gender
ORDER BY Attrition_percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Gender	Total_Employees	Employees_left	Attrition_Percentage
Male	882	150	17.01
Female	588	87	14.80

Business Insight: Male employees (17.01%) have a higher attrition rate than **female employees (14.80%)**, indicating a need for targeted retention strategies.

Overtime vs Employee Attrition Analysis

How does overtime impact employee attrition?

SQL QUERY:

```sql
-- How does overtime impact employee attrition?
SELECT OverTime,
	COUNT(EmployeeNumber) AS Total_employees,
	SUM(
	CASE
	WHEN Attrition="YES" THEN 1
	ELSE 0
	END) AS Employees_left,
	ROUND(
		SUM(
		CASE
		WHEN Attrition="Yes" THEN 1
		ELSE 0
		END)*100.0
		/COUNT(EmployeeNumber),
		2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY Overtime
ORDER BY Attrition_percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OverTime	Total_employees	Employees_left	Attrition_Percentage
Yes	416	127	30.53
No	1054	110	10.44

Business Insight: Employees working **overtime (30.53%)** have a significantly higher attrition rate than those **not working overtime (10.44%)**, indicating that excessive workload may be a key driver of employee turnover.

Salary Slab-wise Employee Attrition Analysis

Which salary slab has the highest employee attrition percentage?

SQL QUERY:

```sql
-- Which salary slab has the highest employee attrition percentage?
SELECT min(MonthlyIncome) AS Min_salary,
       MAX(MonthlyIncome) AS Max_salary
FROM employee_attrition;

SELECT
       CASE
       WHEN MonthlyIncome BETWEEN 1000 AND 5000 THEN "Low Salary"
       WHEN MonthlyIncome BETWEEN 5001 AND 10000 THEN "Medium Salary"
       WHEN MonthlyIncome BETWEEN 10001 AND 15000 THEN "High Salary"
       ELSE "Very High Salary"
       END AS Salary_slabs,
       COUNT(EmployeeNumber) AS Total_employees,
       SUM(
           CASE
           WHEN Attrition="Yes" THEN 1
           ELSE 0
           END) AS Employees_Left,
       ROUND(
             SUM(
                 CASE
           WHEN Attrition="Yes" THEN 1
           ELSE 0
           END)*100.0
           /COUNT(EmployeeNumber),
           2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY Salary_slabs
ORDER BY Attrition_Percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salary_slabs	Total_employees	Employees_Left	Attrition_Percentage
Low Salary	749	163	21.76
High Salary	148	20	13.51
Medium Salary	440	49	11.14
Very High Salary	133	5	3.76

Business Insight: Employees in the **Low Salary** slab (**21.76%**) have the highest attrition rate, while the **Very High Salary** slab (**3.76%**) has the lowest, suggesting that competitive compensation plays a significant role in improving employee retention.

Years at Company vs Employee Attrition Analysis

How does employee attrition vary based on years spent at the company?

SQL QUERY:

```sql
-- How does employee attrition vary based on years spent at the company?
SELECT
        CASE
         WHEN YearsAtCompany BETWEEN 0 AND 2 THEN "New Employee"
         WHEN YearsAtCompany BETWEEN 3 AND 5 THEN "Early Career"
         WHEN YearsAtCompany BETWEEN 6 AND 10 THEN "Mid Career"
         ELSE "Long Tenure"
         END AS ExperienceGroup,
      count(EmployeeNumber) AS total_employees,
      SUM(
         CASE
         WHEN Attrition ="Yes" THEN 1
         ELSE 0
         END) AS Employees_left,
      ROUND(
          SUM(
             CASE
              WHEN Attrition ="Yes" THEN 1
              ELSE 0
              END)*100.0
              /COUNT(EmployeeNumber),
              2) AS Attrition_Percentage
FROM employee_attrition
GROUP BY ExperienceGroup
ORDER BY Attrition_percentage DESC;
```

QUERY OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ExperienceGroup	total_employees	Employees_left	Attrition_Percentage
New Employee	342	102	29.82
Early Career	434	60	13.82
Mid Career	448	55	12.28
Long Tenure	246	20	8.13

Business Insight: **New Employees (29.82%)** have the highest attrition rate, while **Long Tenure employees (8.13%)** have the lowest, indicating that improving onboarding and early employee engagement can significantly reduce turnover.

Overall Project Conclusion & Recommendations

Key Findings:

- Employee attrition is influenced by age, salary, overtime, experience, education field, and gender.
- **New employees**, **low-salary employees**, and those **working overtime** showed the highest attrition rates.
- Employees aged **18–25** experienced the highest turnover, while long-tenured employees had the lowest.
- Employees from **Human Resources** and **Technical Degree** education fields recorded the highest attrition rates.

Business Recommendations:

- Strengthen **onboarding and mentoring** programs for new employees.
- **Review compensation structures** and career growth opportunities for **lower-salary employees**.
- Reduce excessive **overtime** and improve work-life balance.
- Develop targeted retention initiatives for high-risk employee groups.

Project Outcome

This SQL analysis transformed raw HR data into actionable business insights, enabling HR teams to identify attrition patterns and support data-driven workforce decisions.